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1 Orbitals And Their Symmetry

We use the following ordering of orbitals here (assuming degeneracy of LUMOs and HOMOs):

$$\begin{array}{cc} b_2^* \;\text{—} & \text{—}\; a_2^* \\ a_2 \;\text{—} & \text{—}\; b_2 \end{array}$$

In the following, we will leave out the explicit labeling by orbital symmetry. Each monomer will simply be written as:

$$\begin{array}{cc} \text{—} & \text{—} \\ \text{—} & \text{—} \end{array}$$

where the character of the orbital is defined by its position.

Dimer configurations will be written as their monomer occupations, by writing one monomer to the left and one to the right. The orbital ordering within the group of monomer orbitals follows the above mentioned definition.

The transformation of monomer orbitals into dimer orbitals is given by knowing the negative and positive linear combinations of monomer orbitals into dimer orbitals. Transforming this known equation system leads to:

$$\begin{aligned}
a_2^{*l} &= +a_u^* + b_g^* \\
b_2^{*l} &= +a_g^* + b_u^* \\
a_2^{l} &= +a_u + b_g \\
b_2^{l} &= +a_g + b_u \\
a_2^{*r} &= +a_u^* - b_g^* \\
b_2^{*r} &= -a_g^* + b_u^* \\
a_2^{r} &= +a_u - b_g \\
b_2^{r} &= -a_g + b_u
\end{aligned}$$

2 Monomer States and Configurations

$$S = \quad + \begin{pmatrix} \text{---} & \text{---} \\ \downarrow\uparrow & \downarrow\uparrow \end{pmatrix} \quad = \left| \underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^0}_{a_g} \right|$$

$$Q = \quad + \begin{pmatrix} \uparrow & \uparrow \\ \uparrow & \uparrow \end{pmatrix} \quad = \left| \underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^1}_{a_1} \right|$$

$$i^3a_1 = \quad + \begin{pmatrix} \uparrow & \text{---} \\ \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \text{---} & \uparrow \\ \uparrow & \downarrow\uparrow \end{pmatrix} \quad = \left| \underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1} \right| - \left| \underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1} \right|$$

$$e^3a_1 = \quad + \begin{pmatrix} \uparrow & \text{---} \\ \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \text{---} & \uparrow \\ \uparrow & \downarrow\uparrow \end{pmatrix} \quad = \left| \underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1} \right| + \left| \underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1} \right|$$

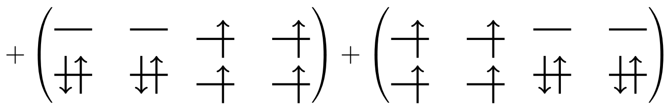
$$i^3b_1 = \quad + \begin{pmatrix} \uparrow & \text{---} \\ \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \text{---} & \uparrow \\ \downarrow\uparrow & \uparrow \end{pmatrix} \quad = \left| \underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1} \right| - \left| \underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1} \right|$$

$$e^3b_1 = \quad + \begin{pmatrix} \uparrow & \text{---} \\ \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \text{---} & \uparrow \\ \downarrow\uparrow & \uparrow \end{pmatrix} \quad = \left| \underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1} \right| + \left| \underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1} \right|$$

3 Linear Combinations

3.1 $S * Q + Q * S$

$$S * Q + Q * S$$



monomer ci vectors:

$$+\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^0}_{a_g^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right| + \left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^0}_{a_g^r}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+a_g+b_u)^2(+a_g^*+b_u^*)^0(+a_u+b_g)^2(+a_u^*+b_g^*)^0}_{a_g^l}\right| \cdot \left|\underbrace{(-a_g+b_u)^1(-a_g^*+b_u^*)^1(+a_u-b_g)^1(+a_u^*-b_g^*)^1}_{a_1^r}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^1(b_g^*)^1}_{a_g}\right| + 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^0}_{a_g}\right|$$

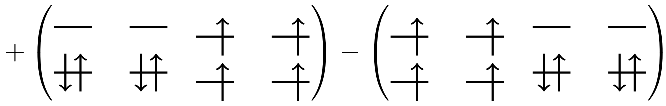
$$-2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^1(b_g^*)^0}_{a_g}\right| + 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^1}_{a_g}\right|$$

$$+2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^0}_{a_g}\right| - 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^2(b_g^*)^1}_{a_g}\right|$$

$$+2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^1}_{a_g}\right| + 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^2(b_g^*)^0}_{a_g}\right|$$

3.2 $S * Q - Q * S$

$$S * Q - Q * S$$



monomer ci vectors:

$$+\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^0}_{a_g^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right| - \left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^0}_{a_g^r}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+a_g+b_u)^2(+a_g^*+b_u^*)^0(+a_u+b_g)^2(+a_u^*+b_g^*)^0}_{a_g^l}\right| \cdot \left|\underbrace{(-a_g+b_u)^1(-a_g^*+b_u^*)^1(+a_u-b_g)^1(+a_u^*-b_g^*)^1}_{a_1^r}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^1(b_g)^1(b_g^*)^0}_{b_u}\right| - 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^1}_{b_u}\right|$$

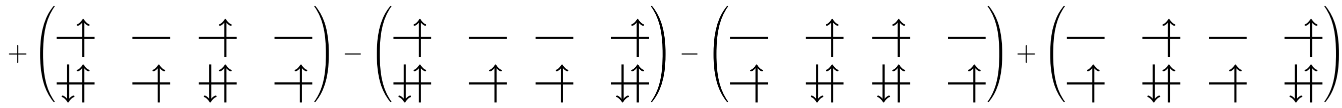
$$+2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^0(b_g)^1(b_g^*)^1}_{b_u}\right| + 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^0}_{b_u}\right|$$

$$-2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^1}_{b_u}\right| - 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^1(b_g)^2(b_g^*)^0}_{b_u}\right|$$

$$+2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^0}_{b_u}\right| - 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^0(b_g)^2(b_g^*)^1}_{b_u}\right|$$

3.3 $i^3a_1 * i^3a_1$

$i^3a_1 * i^3a_1$



monomer ci vectors:

$$+\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right|\cdot\left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right| - \left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right|\cdot\left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$

$$-\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right|\cdot\left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right| + \left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right|\cdot\left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+a_g+b_u)^1(+a_g^*+b_u^*)^1(+a_u+b_g)^2(+a_u^*+b_g^*)^0}_{a_1^l}\right|\cdot\left|\underbrace{(-a_g+b_u)^1(-a_g^*+b_u^*)^1(+a_u-b_g)^2(+a_u^*-b_g^*)^0}_{a_1^r}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^0}_{a_g}\right| - 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^1}_{a_g}\right|$$

$$-2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^2(b_g^*)^0}_{a_g}\right| + 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^0}_{a_g}\right|$$

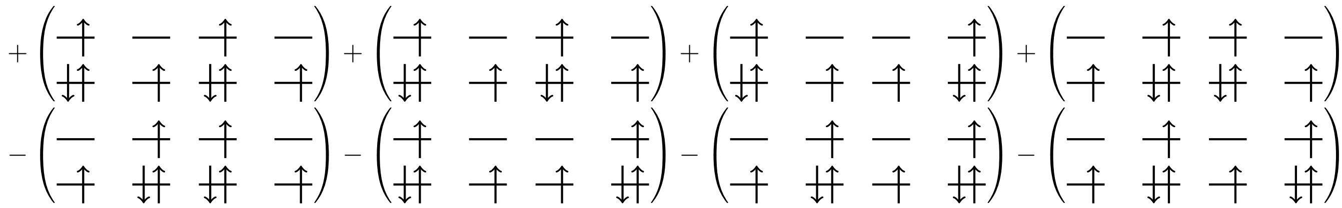
$$-2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^2(b_g^*)^1}_{a_g}\right| - 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^1(b_g^*)^0}_{a_g}\right|$$

$$+2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^1}_{a_g}\right| - 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^1(b_g^*)^1}_{a_g}\right|$$

$$-2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^0}_{a_g}\right| + \left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^1}_{a_g}\right|$$

3.4 $i^3a_1 * e^3a_1 + e^3a_1 * i^3a_1$

$$i^3a_1 * e^3a_1 + e^3a_1 * i^3a_1$$



monomer ci vectors:

$$
\begin{aligned}
&+\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right| + \left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right| \\
&+\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right| + \left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right| \\
&-\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right| - \left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right| \\
&-\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right| - \left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|
\end{aligned}
$$

substitution by dimer orbitals:

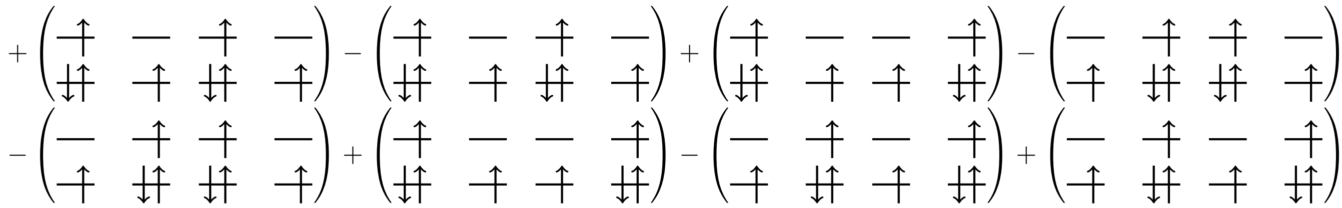
$$+\left|\underbrace{(+a_g+b_u)^1(+a_g^*+b_u^*)^1(+a_u+b_g)^2(+a_u^*+b_g^*)^0}_{a_1^l}\right| \cdot \left|\underbrace{(-a_g+b_u)^1(-a_g^*+b_u^*)^1(+a_u-b_g)^2(+a_u^*-b_g^*)^0}_{a_1^r}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^0}_{a_g}\right| - 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^1}_{a_g}\right|$$

3.5 $i^3a_1 * e^3a_1 - e^3a_1 * i^3a_1$

$$i^3a_1 * e^3a_1 - e^3a_1 * i^3a_1$$



monomer ci vectors:

$$+\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right| - \left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right|$$

$$+\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right| - \left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right|$$

$$-\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right| + \left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$

$$-\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right| + \left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+a_g+b_u)^1(+a_g^*+b_u^*)^1(+a_u+b_g)^2(+a_u^*+b_g^*)^0}_{a_1^l}\right| \cdot \left|\underbrace{(-a_g+b_u)^1(-a_g^*+b_u^*)^1(+a_u-b_g)^2(+a_u^*-b_g^*)^0}_{a_1^r}\right| + \ldots$$

(multiplied out and summed up) dimer ci vectors:

$$+4\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^0}_{b_u}\right| - 4\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^0(b_g)^2(b_g^*)^1}_{b_u}\right|$$

$$+4\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^1}_{b_u}\right| + 4\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^1(b_g)^2(b_g^*)^0}_{b_u}\right|$$

$$-4\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^0(b_g)^1(b_g^*)^1}_{b_u}\right| - 4\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^0}_{b_u}\right|$$

$$+4\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^1(b_g)^1(b_g^*)^0}_{b_u}\right| - 4\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^1}_{b_u}\right|$$

3.6 $i^3a_1 * i^3b_1 + i^3b_1 * i^3a_1$

$$i^3a_1 * i^3b_1 + i^3b_1 * i^3a_1$$

$$+\begin{pmatrix} \uparrow & \text{—} & \uparrow & \text{—} \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & \text{—} & \uparrow & \text{—} \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & \text{—} & \text{—} & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \text{—} & \uparrow & \uparrow & \text{—} \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$
$$-\begin{pmatrix} \text{—} & \uparrow & \uparrow & \text{—} \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & \text{—} & \text{—} & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \text{—} & \uparrow & \text{—} & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \text{—} & \uparrow & \text{—} & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

monomer ci vectors:

$$+\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right|$$
$$-\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right| - \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right|$$
$$-\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| - \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$
$$+\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+a_g+b_u)^1(+a_g^*+b_u^*)^1(+a_u+b_g)^2(+a_u^*+b_g^*)^0}_{a_1^l}\right| \cdot \left|\underbrace{(-a_g+b_u)^2(-a_g^*+b_u^*)^1(+a_u-b_g)^1(+a_u^*-b_g^*)^0}_{b_1^r}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^2(b_g^*)^0}_{b_g}\right| - 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^1(b_g^*)^0}_{b_g}\right|$$
$$-2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^0}_{b_g}\right| + 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^1}_{b_g}\right|$$
$$+2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^1}_{b_g}\right| - 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^0}_{b_g}\right|$$
$$-2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^1(b_g^*)^1}_{b_g}\right| + 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^2(b_g^*)^1}_{b_g}\right|$$

3.7 $i^3a_1 * i^3b_1 - i^3b_1 * i^3a_1$

$$i^3a_1 * i^3b_1 - i^3b_1 * i^3a_1$$

$$+\begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$
$$-\begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

monomer ci vectors:

$$+\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| - \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right|$$
$$-\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right|$$
$$-\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$
$$+\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right| - \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$

substitution by dimer orbitals:

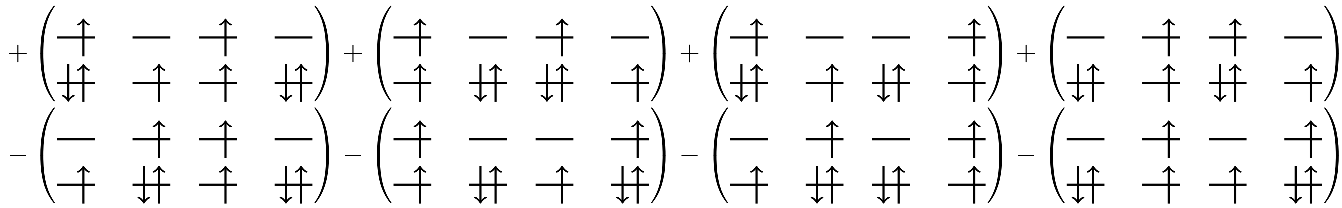
$$+\left|\underbrace{(+a_g+b_u)^1(+a_g^*+b_u^*)^1(+a_u+b_g)^2(+a_u^*+b_g^*)^0}_{a_1^l}\right| \cdot \left|\underbrace{(-a_g+b_u)^2(-a_g^*+b_u^*)^1(+a_u-b_g)^1(+a_u^*-b_g^*)^0}_{b_1^r}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$-2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^0}_{a_u}\right| - 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^0}_{a_u}\right|$$
$$+2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^1}_{a_u}\right| + 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^0}_{a_u}\right|$$
$$-2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^0}_{a_u}\right| - 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^1}_{a_u}\right|$$
$$+2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^1}_{a_u}\right| + 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^1}_{a_u}\right|$$

3.8 $i^3a_1 * e^3b_1 + e^3b_1 * i^3a_1$

$$i^3a_1 * e^3b_1 + e^3b_1 * i^3a_1$$



monomer ci vectors:

$$+\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right|$$

$$+\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right|$$

$$-\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| - \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$

$$-\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right| - \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+a_g+b_u)^1(+a_g^*+b_u^*)^1(+a_u+b_g)^2(+a_u^*+b_g^*)^0}_{a_1^l}\right| \cdot \left|\underbrace{(-a_g+b_u)^2(-a_g^*+b_u^*)^1(+a_u-b_g)^1(+a_u^*-b_g^*)^0}_{b_1^r}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^2(b_g^*)^0}_{b_g}\right| - 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^1(b_g^*)^0}_{b_g}\right|$$

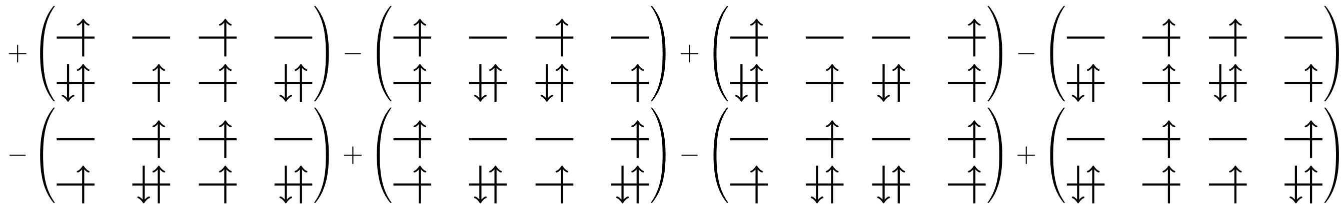
$$+2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^0}_{b_g}\right| - 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^1}_{b_g}\right|$$

$$+2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^1}_{b_g}\right| - 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^0}_{b_g}\right|$$

$$+2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^1(b_g^*)^1}_{b_g}\right| - 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^2(b_g^*)^1}_{b_g}\right|$$

3.9 $i^3a_1 * e^3b_1 - e^3b_1 * i^3a_1$

$$i^3a_1 * e^3b_1 - e^3b_1 * i^3a_1$$



monomer ci vectors:

$$+\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| - \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right|$$

$$+\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right| - \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right|$$

$$-\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$

$$-\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+a_g+b_u)^1(+a_g^*+b_u^*)^1(+a_u+b_g)^2(+a_u^*+b_g^*)^0}_{a_1^l}\right| \cdot \left|\underbrace{(-a_g+b_u)^2(-a_g^*+b_u^*)^1(+a_u-b_g)^1(+a_u^*-b_g^*)^0}_{b_1^r}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$-2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^0}_{a_u}\right| - 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^0}_{a_u}\right|$$

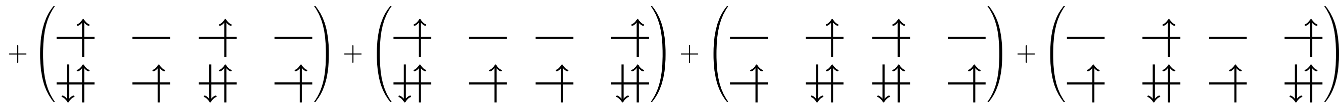
$$-2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^1}_{a_u}\right| - 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^0}_{a_u}\right|$$

$$-2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^0}_{a_u}\right| - 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^1}_{a_u}\right|$$

$$-2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^1}_{a_u}\right| - 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^1}_{a_u}\right|$$

3.10 $e^3a_1 * e^3a_1$

$e^3a_1 * e^3a_1$



monomer ci vectors:

$$+\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right| + \left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$
$$+\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right| + \left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+a_g+b_u)^1(+a_g^*+b_u^*)^1(+a_u+b_g)^2(+a_u^*+b_g^*)^0}_{a_1^l}\right| \cdot \left|\underbrace{(-a_g+b_u)^1(-a_g^*+b_u^*)^1(+a_u-b_g)^2(+a_u^*-b_g^*)^0}_{a_1^r}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^0}_{a_g}\right| + 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^1}_{a_g}\right|$$
$$+2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^2(b_g^*)^0}_{a_g}\right| - 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^0}_{a_g}\right|$$
$$+2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^2(b_g^*)^1}_{a_g}\right| + 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^1(b_g^*)^0}_{a_g}\right|$$
$$-2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^1}_{a_g}\right| + 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^1(b_g^*)^1}_{a_g}\right|$$
$$+2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^0}_{a_g}\right| + \left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^1}_{a_g}\right|$$

3.11 $e^3a_1 * i^3b_1 + i^3b_1 * e^3a_1$

$$e^3a_1 * i^3b_1 + i^3b_1 * e^3a_1$$

$$+\begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$
$$+\begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

monomer ci vectors:

$$+\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right|$$
$$-\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right| - \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right|$$
$$+\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$
$$-\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right| - \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$

substitution by dimer orbitals:

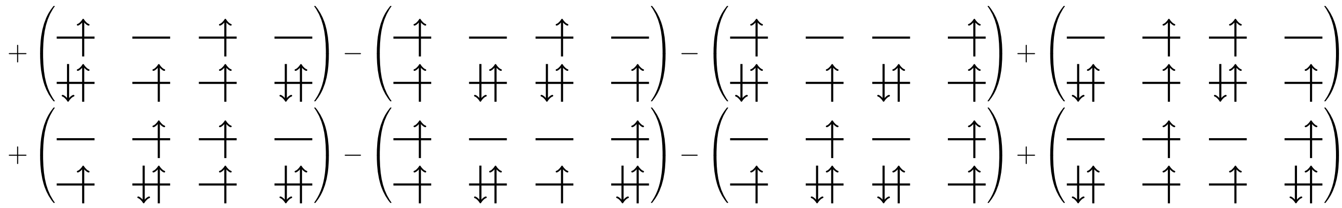
$$+\left|\underbrace{(+a_g+b_u)^1(+a_g^*+b_u^*)^1(+a_u+b_g)^2(+a_u^*+b_g^*)^0}_{a_1^l}\right| \cdot \left|\underbrace{(-a_g+b_u)^2(-a_g^*+b_u^*)^1(+a_u-b_g)^1(+a_u^*-b_g^*)^0}_{b_1^r}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^2(b_g^*)^0}_{b_g}\right| - 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^1(b_g^*)^0}_{b_g}\right|$$
$$-2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^0}_{b_g}\right| + 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^1}_{b_g}\right|$$
$$-2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^1}_{b_g}\right| + 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^0}_{b_g}\right|$$
$$+2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^1(b_g^*)^1}_{b_g}\right| - 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^2(b_g^*)^1}_{b_g}\right|$$

3.12 $e^3a_1 * i^3b_1 - i^3b_1 * e^3a_1$

$$e^3a_1 * i^3b_1 - i^3b_1 * e^3a_1$$



monomer ci vectors:

$$
\begin{aligned}
&+\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right|\cdot\left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| - \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right|\cdot\left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right|\\
&-\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right|\cdot\left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right|\cdot\left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right|\\
&+\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right|\cdot\left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| - \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right|\cdot\left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|\\
&-\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right|\cdot\left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right|\cdot\left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|
\end{aligned}
$$

substitution by dimer orbitals:

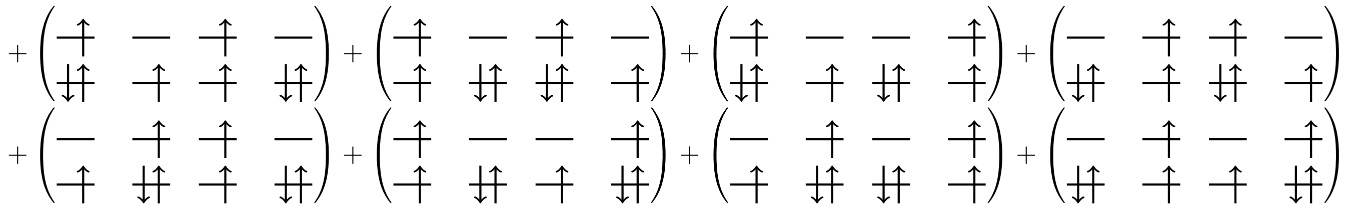
$$+\left|\underbrace{(+a_g+b_u)^1(+a_g^*+b_u^*)^1(+a_u+b_g)^2(+a_u^*+b_g^*)^0}_{a_1^l}\right|\cdot\left|\underbrace{(-a_g+b_u)^2(-a_g^*+b_u^*)^1(+a_u-b_g)^1(+a_u^*-b_g^*)^0}_{b_1^r}\right|+\dots$$

(multiplied out and summed up) dimer ci vectors:

$$
\begin{aligned}
&-2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^0}_{a_u}\right| - 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^0}_{a_u}\right|\\
&+2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^1}_{a_u}\right| + 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^0}_{a_u}\right|\\
&+2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^0}_{a_u}\right| + 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^1}_{a_u}\right|\\
&-2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^1}_{a_u}\right| - 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^1}_{a_u}\right|
\end{aligned}
$$

3.13 $e^3a_1 * e^3b_1 + e^3b_1 * e^3a_1$

$$e^3a_1 * e^3b_1 + e^3b_1 * e^3a_1$$



monomer ci vectors:

$$+\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right|$$
$$+\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right|$$
$$+\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$
$$+\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$

substitution by dimer orbitals:

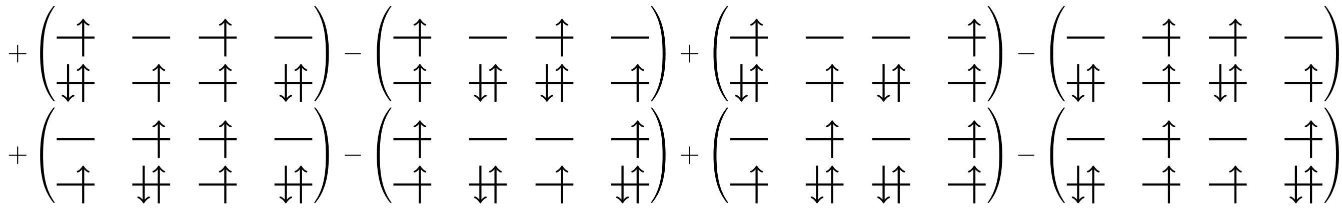
$$+\left|\underbrace{(+a_g+b_u)^1(+a_g^*+b_u^*)^1(+a_u+b_g)^2(+a_u^*+b_g^*)^0}_{a_1^l}\right| \cdot \left|\underbrace{(-a_g+b_u)^2(-a_g^*+b_u^*)^1(+a_u-b_g)^1(+a_u^*-b_g^*)^0}_{b_1^r}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^2(b_g^*)^0}_{b_g}\right| - 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^1(b_g^*)^0}_{b_g}\right|$$
$$+2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^0}_{b_g}\right| - 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^1}_{b_g}\right|$$
$$-2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^1}_{b_g}\right| + 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^0}_{b_g}\right|$$
$$-2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^1(b_g^*)^1}_{b_g}\right| + 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^2(b_g^*)^1}_{b_g}\right|$$

3.14 $e^3a_1 * e^3b_1 - e^3b_1 * e^3a_1$

$$e^3a_1 * e^3b_1 - e^3b_1 * e^3a_1$$



monomer ci vectors:

$$+\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| - \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right|$$

$$+\left|\underbrace{(b_2^l)^1(b_2^{*l})^1(a_2^l)^2(a_2^{*l})^0}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right| - \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^1(a_2^r)^2(a_2^{*r})^0}_{a_1^r}\right|$$

$$+\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| - \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$

$$+\left|\underbrace{(b_2^l)^2(b_2^{*l})^0(a_2^l)^1(a_2^{*l})^1}_{a_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right| - \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^0(a_2^r)^1(a_2^{*r})^1}_{a_1^r}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+a_g+b_u)^1(+a_g^*+b_u^*)^1(+a_u+b_g)^2(+a_u^*+b_g^*)^0}_{a_1^l}\right| \cdot \left|\underbrace{(-a_g+b_u)^2(-a_g^*+b_u^*)^1(+a_u-b_g)^1(+a_u^*-b_g^*)^0}_{b_1^r}\right| + \ldots$$

(multiplied out and summed up) dimer ci vectors:

$$-2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^0}_{a_u}\right| - 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^0}_{a_u}\right|$$

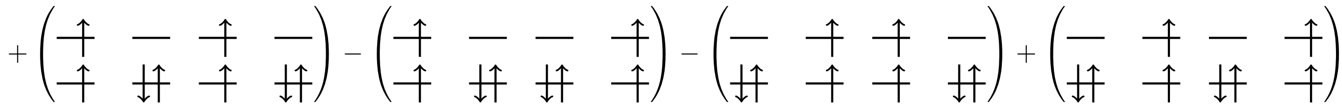
$$-2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^1}_{a_u}\right| - 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^0}_{a_u}\right|$$

$$+2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^0}_{a_u}\right| + 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^1}_{a_u}\right|$$

$$+2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^1}_{a_u}\right| + 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^1}_{a_u}\right|$$

3.15 $i^3 b_1 * i^3 b_1$

$i^3 b_1 * i^3 b_1$



monomer ci vectors:

$$
\begin{aligned}
&+\left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| - \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right| \\
&-\left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right|
\end{aligned}
$$

substitution by dimer orbitals:

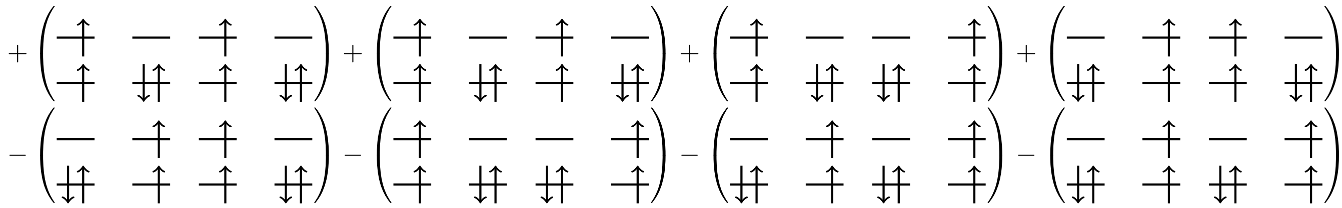
$$
+\left|\underbrace{(+a_g+b_u)^2(+a_g^*+b_u^*)^1(+a_u+b_g)^1(+a_u^*+b_g^*)^0}_{b_1^l}\right| \cdot \left|\underbrace{(-a_g+b_u)^2(-a_g^*+b_u^*)^1(+a_u-b_g)^1(+a_u^*-b_g^*)^0}_{b_1^r}\right| + \dots
$$

(multiplied out and summed up) dimer ci vectors:

$$
\begin{aligned}
&+\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^0}_{a_g}\right| + 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^1}_{a_g}\right| \\
&+2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^1(b_g^*)^0}_{a_g}\right| - 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^0}_{a_g}\right| \\
&+2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^1(b_g^*)^1}_{a_g}\right| + 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^2(b_g^*)^0}_{a_g}\right| \\
&-2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^1}_{a_g}\right| + 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^2(b_g^*)^1}_{a_g}\right| \\
&+2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^0}_{a_g}\right| + \left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^1}_{a_g}\right|
\end{aligned}
$$

3.16 $i^3b_1 * e^3b_1 + e^3b_1 * i^3b_1$

$$i^3b_1 * e^3b_1 + e^3b_1 * i^3b_1$$



monomer ci vectors:

$$
\begin{aligned}
&+\left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right|\cdot\left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right|+\left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right|\cdot\left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right|\\
&+\left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right|\cdot\left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right|+\left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right|\cdot\left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right|\\
&-\left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right|\cdot\left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right|-\left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right|\cdot\left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right|\\
&-\left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right|\cdot\left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right|-\left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right|\cdot\left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right|
\end{aligned}
$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+a_g+b_u)^2(+a_g^*+b_u^*)^1(+a_u+b_g)^1(+a_u^*+b_g^*)^0}_{b_1^l}\right|\cdot\left|\underbrace{(-a_g+b_u)^2(-a_g^*+b_u^*)^1(+a_u-b_g)^1(+a_u^*-b_g^*)^0}_{b_1^r}\right|+\dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^0}_{a_g}\right|-2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^1}_{a_g}\right|$$

3.17 $i^3b_1 * e^3b_1 - e^3b_1 * i^3b_1$

$$i^3b_1 * e^3b_1 - e^3b_1 * i^3b_1$$

$$+\begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$
$$-\begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

monomer ci vectors:

$$+\left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| - \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right|$$
$$+\left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right| - \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right|$$
$$-\left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right|$$
$$-\left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right|$$

substitution by dimer orbitals:

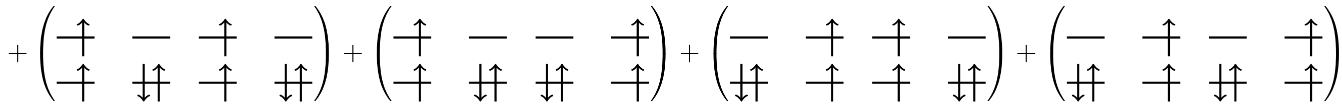
$$+\left|\underbrace{(+a_g+b_u)^2(+a_g^*+b_u^*)^1(+a_u+b_g)^1(+a_u^*+b_g^*)^0}_{b_1^l}\right| \cdot \left|\underbrace{(-a_g+b_u)^2(-a_g^*+b_u^*)^1(+a_u-b_g)^1(+a_u^*-b_g^*)^0}_{b_1^r}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$-4 \cdot \left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^0}_{b_u}\right| + 4 \cdot \left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^0(b_g)^1(b_g^*)^1}_{b_u}\right|$$
$$-4 \cdot \left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^1}_{b_u}\right| - 4 \cdot \left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^1(b_g)^1(b_g^*)^0}_{b_u}\right|$$
$$+4 \cdot \left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^0(b_g)^2(b_g^*)^1}_{b_u}\right| + 4 \cdot \left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^0}_{b_u}\right|$$
$$-4 \cdot \left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^1(b_g)^2(b_g^*)^0}_{b_u}\right| + 4 \cdot \left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^1}_{b_u}\right|$$

3.18 $e^3b_1 * e^3b_1$

$$e^3b_1 * e^3b_1$$



monomer ci vectors:

$$+\left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^2(b_2^{*l})^1(a_2^l)^1(a_2^{*l})^0}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right|$$
$$+\left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^2(b_2^{*r})^1(a_2^r)^1(a_2^{*r})^0}_{b_1^r}\right| + \left|\underbrace{(b_2^l)^1(b_2^{*l})^0(a_2^l)^2(a_2^{*l})^1}_{b_1^l}\right| \cdot \left|\underbrace{(b_2^r)^1(b_2^{*r})^0(a_2^r)^2(a_2^{*r})^1}_{b_1^r}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+a_g+b_u)^2(+a_g^*+b_u^*)^1(+a_u+b_g)^1(+a_u^*+b_g^*)^0}_{b_1^l}\right| \cdot \left|\underbrace{(-a_g+b_u)^2(-a_g^*+b_u^*)^1(+a_u-b_g)^1(+a_u^*-b_g^*)^0}_{b_1^r}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^0}_{a_g}\right| - 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^1(a_u^*)^0(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^1}_{a_g}\right|$$
$$-2\cdot\left|\underbrace{(a_g)^2(a_g^*)^1(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^1(b_g^*)^0}_{a_g}\right| + 2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^1(a_u^*)^1(b_u)^1(b_u^*)^1(b_g)^2(b_g^*)^0}_{a_g}\right|$$
$$-2\cdot\left|\underbrace{(a_g)^2(a_g^*)^0(a_u)^2(a_u^*)^0(b_u)^1(b_u^*)^1(b_g)^1(b_g^*)^1}_{a_g}\right| - 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^1(a_u^*)^1(b_u)^2(b_u^*)^0(b_g)^2(b_g^*)^0}_{a_g}\right|$$
$$+2\cdot\left|\underbrace{(a_g)^1(a_g^*)^1(a_u)^2(a_u^*)^0(b_u)^2(b_u^*)^0(b_g)^1(b_g^*)^1}_{a_g}\right| - 2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^1(a_u^*)^0(b_u)^2(b_u^*)^1(b_g)^2(b_g^*)^1}_{a_g}\right|$$
$$-2\cdot\left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^2(b_u^*)^1(b_g)^1(b_g^*)^0}_{a_g}\right| + \left|\underbrace{(a_g)^1(a_g^*)^0(a_u)^2(a_u^*)^1(b_u)^1(b_u^*)^0(b_g)^2(b_g^*)^1}_{a_g}\right|$$

4 Symmetry Conclusion

Dimer State	Symmetry
$S * Q + Q * S$	a_g
$S * Q - Q * S$	b_u
$i^3a_1 * i^3a_1$	a_g
$i^3a_1 * e^3a_1 + e^3a_1 * i^3a_1$	a_g
$i^3a_1 * e^3a_1 - e^3a_1 * i^3a_1$	b_u
$i^3a_1 * i^3b_1 + i^3b_1 * i^3a_1$	b_g
$i^3a_1 * i^3b_1 - i^3b_1 * i^3a_1$	a_u
$i^3a_1 * e^3b_1 + e^3b_1 * i^3a_1$	b_g
$i^3a_1 * e^3b_1 - e^3b_1 * i^3a_1$	a_u
$e^3a_1 * e^3a_1$	a_g
$e^3a_1 * i^3b_1 + i^3b_1 * e^3a_1$	b_g
$e^3a_1 * i^3b_1 - i^3b_1 * e^3a_1$	a_u
$e^3a_1 * e^3b_1 + e^3b_1 * e^3a_1$	b_g
$e^3a_1 * e^3b_1 - e^3b_1 * e^3a_1$	a_u
$i^3b_1 * i^3b_1$	a_g
$i^3b_1 * e^3b_1 + e^3b_1 * i^3b_1$	a_g
$i^3b_1 * e^3b_1 - e^3b_1 * i^3b_1$	b_u
$e^3b_1 * e^3b_1$	a_g

$\Rightarrow$ 7x a_g and 3x b_u and 4x b_g and 4x a_u

5 CI Vector Conclusion

Dimer States and Their CI Vectors:

- $\boldsymbol{S * Q + Q * S}$:

$$+2 \cdot |2020aaaa| + 2 \cdot |20aaaa20| - 2 \cdot |2a2aa0a0| + 2 \cdot |2aa0a02a| + 2 \cdot |a02a2aa0| - 2 \cdot |a0a02a2a|$$

$$+2 \cdot |aa2020aa| + 2 \cdot |aaaa2020|$$

- $\boldsymbol{S * Q - Q * S}$:

$$+2 \cdot |202aaaa0| - 2 \cdot |20a0aa2a| + 2 \cdot |2a20a0aa| + 2 \cdot |2aaaa020| - 2 \cdot |a0202aaa| - 2 \cdot |a0aa2a20|$$

$$+2 \cdot |aa2a20a0| - 2 \cdot |aaa0202a|$$

- $\boldsymbol{i^3a_1 * i^3a_1}$:

$$+ |aa20aa20| - 2 \cdot |aa2020aa| - 2 \cdot |aaaa2020| + 2 \cdot |a02a2aa0| - 2 \cdot |a0a02a2a| - 2 \cdot |2a2aa0a0|$$

$$+2 \cdot |2aa0a02a| - 2 \cdot |2020aaaa| - 2 \cdot |20aaaa20| + |20aa20aa|$$

- $\boldsymbol{i^3a_1 * e^3a_1 + e^3a_1 * i^3a_1}$:

$$+2 \cdot |aa20aa20| - 2 \cdot |20aa20aa|$$

- $\boldsymbol{i^3a_1 * e^3a_1 - e^3a_1 * i^3a_1}$:

$$+4 \cdot |aa2a20a0| - 4 \cdot |aaa0202a| + 4 \cdot |a0202aaa| + 4 \cdot |a0aa2a20| - 4 \cdot |2a20a0aa| - 4 \cdot |2aaaa020|$$

$$+4 \cdot |202aaaa0| - 4 \cdot |20a0aa2a|$$

- $\boldsymbol{i^3a_1 * i^3b_1 + i^3b_1 * i^3a_1}$:

$$+2 \cdot |aaa02a20| - 2 \cdot |2a20aaa0| - 2 \cdot |aa2aa020| + 2 \cdot |a020aa2a| + 2 \cdot |20a02aaa| - 2 \cdot |2aaa20a0|$$

$$-2 \cdot |202aa0aa| + 2 \cdot |a0aa202a|$$

- $\boldsymbol{i^3a_1 * i^3b_1 - i^3b_1 * i^3a_1}$:

$$-2 \cdot |aa202aa0| - 2 \cdot |2aa0aa20| + 2 \cdot |aa20a02a| + 2 \cdot |a02aaa20| - 2 \cdot |20aa2aa0| - 2 \cdot |2aa020aa|$$

$$+2 \cdot |20aaa02a| + 2 \cdot |a02a20aa|$$

- $\boldsymbol{i^3a_1 * e^3b_1 + e^3b_1 * i^3a_1}$:

$$+2 \cdot |aaa02a20| - 2 \cdot |2a20aaa0| + 2 \cdot |aa2aa020| - 2 \cdot |a020aa2a| + 2 \cdot |20a02aaa| - 2 \cdot |2aaa20a0|$$

$$+2 \cdot |202aa0aa| - 2 \cdot |a0aa202a|$$

- $\boldsymbol{i^3a_1 * e^3b_1 - e^3b_1 * i^3a_1}$:

$$-2 \cdot |aa202aa0| - 2 \cdot |2aa0aa20| - 2 \cdot |aa20a02a| - 2 \cdot |a02aaa20| - 2 \cdot |20aa2aa0| - 2 \cdot |2aa020aa|$$

$$-2 \cdot |20aaa02a| - 2 \cdot |a02a20aa|$$

- $\boldsymbol{e^3a_1 * e^3a_1}$:

$$+ |aa20aa20| + 2 \cdot |aa2020aa| + 2 \cdot |aaaa2020| - 2 \cdot |a02a2aa0| + 2 \cdot |a0a02a2a| + 2 \cdot |2a2aa0a0|$$

$$-2 \cdot |2aa0a02a| + 2 \cdot |2020aaaa| + 2 \cdot |20aaaa20| + |20aa20aa|$$

- $\boldsymbol{e^3a_1 * i^3b_1 + i^3b_1 * e^3a_1}$:

$$+2\cdot|aaa02a20| - 2\cdot|2a20aaa0| - 2\cdot|aa2aa020| + 2\cdot|a020aa2a| - 2\cdot|20a02aaa| + 2\cdot|2aaa20a0|$$

$$+2\cdot|202aa0aa| - 2\cdot|a0aa202a|$$

- $\boldsymbol{e^3a_1 * i^3b_1 - i^3b_1 * e^3a_1}$:

$$-2\cdot|aa202aa0| - 2\cdot|2aa0aa20| + 2\cdot|aa20a02a| + 2\cdot|a02aaa20| + 2\cdot|20aa2aa0| + 2\cdot|2aa020aa|$$

$$-2\cdot|20aaa02a| - 2\cdot|a02a20aa|$$

- $\boldsymbol{e^3a_1 * e^3b_1 + e^3b_1 * e^3a_1}$:

$$+2\cdot|aaa02a20| - 2\cdot|2a20aaa0| + 2\cdot|aa2aa020| - 2\cdot|a020aa2a| - 2\cdot|20a02aaa| + 2\cdot|2aaa20a0|$$

$$-2\cdot|202aa0aa| + 2\cdot|a0aa202a|$$

- $\boldsymbol{e^3a_1 * e^3b_1 - e^3b_1 * e^3a_1}$:

$$-2\cdot|aa202aa0| - 2\cdot|2aa0aa20| - 2\cdot|aa20a02a| - 2\cdot|a02aaa20| + 2\cdot|20aa2aa0| + 2\cdot|2aa020aa|$$

$$+2\cdot|20aaa02a| + 2\cdot|a02a20aa|$$

- $\boldsymbol{i^3b_1 * i^3b_1}$:

$$+|2aa02aa0| + 2\cdot|2aa0a02a| + 2\cdot|2a2aa0a0| - 2\cdot|20aaaa20| + 2\cdot|2020aaaa| + 2\cdot|aaaa2020|$$

$$-2\cdot|aa2020aa| + 2\cdot|a0a02a2a| + 2\cdot|a02a2aa0| + |a02aa02a|$$

- $\boldsymbol{i^3b_1 * e^3b_1 + e^3b_1 * i^3b_1}$:

$$+2\cdot|2aa02aa0| - 2\cdot|a02aa02a|$$

- $\boldsymbol{i^3b_1 * e^3b_1 - e^3b_1 * i^3b_1}$:

$$-4\cdot|2aaaa020| + 4\cdot|2a20a0aa| - 4\cdot|20a0aa2a| - 4\cdot|202aaaa0| + 4\cdot|aaa0202a| + 4\cdot|aa2a20a0|$$

$$-4\cdot|a0aa2a20| + 4\cdot|a0202aaa|$$

- $\boldsymbol{e^3b_1 * e^3b_1}$:

$$+|2aa02aa0| - 2\cdot|2aa0a02a| - 2\cdot|2a2aa0a0| + 2\cdot|20aaaa20| - 2\cdot|2020aaaa| - 2\cdot|aaaa2020|$$

$$+2\cdot|aa2020aa| - 2\cdot|a0a02a2a| - 2\cdot|a02a2aa0| + |a02aa02a|$$